

Math 110 Proof Playbook

This is **not** a list of past problems — it's a set of reusable *templates* for the kinds of arguments this course keeps asking for. When you hit a proof question on the final, scan the "trigger" lines below to figure out which template(s) apply, then follow the steps.

1. Proving $V = U \oplus W$ (direct sum decomposition)

Trigger: "show V is the direct sum of...", "show every v decomposes uniquely as..."

Method A (sum + trivial intersection) — fastest when U, W are subspaces you can describe directly:

1. Show $U + W = V$: take arbitrary $v \in V$, exhibit $u \in U, w \in W$ with $v = u + w$ (often via an explicit formula, e.g. $v = Tv + (v - Tv)$).
2. Show $U \cap W = \{0\}$: let $x \in U \cap W$, use properties of U and W to force $x = 0$.
3. Conclude $V = U \oplus W$.

Method B (dimension counting) — fastest once you already know $U + W = V$ or $U \cap W = \{0\}$:

- If $\dim U + \dim W = \dim V$ and $U \cap W = \{0\}$, direct sum follows automatically (no need to separately check $U + W = V$).
- If $\dim U + \dim W = \dim V$ and $U + W = V$, direct sum follows automatically (no need to separately check trivial intersection).

Method C (via rank-nullity, for $V = \text{null}(T) \oplus \text{range}(T)$ -type claims):

- Use rank-nullity to get the dimension count for free, then just show trivial intersection.

Warning: for 3+ subspaces, pairwise trivial intersection is **not enough** — you need the Gamma-map injectivity condition (or the dimension-sum condition) instead.

2. Proving something is a subspace

Trigger: "show U is a subspace of..."

1. $0 \in U$.
2. Closed under addition: $u_1, u_2 \in U \Rightarrow u_1 + u_2 \in U$.
3. Closed under scalar multiplication: $u \in U, c \in F \Rightarrow cu \in U$.

(Steps 2–3 can be combined into one "closed under linear combinations" check.)

3. Proving a linear map is injective / surjective / invertible

Trigger: "show T is injective/surjective/invertible", "show T is unique" (uniqueness of linear maps satisfying conditions on a basis)

- **Injective:** show $\text{null}(T) = \{0\}$ — assume $Tv = 0$, derive $v = 0$.
 - **Surjective:** show $\text{range}(T) = W$ — given arbitrary $w \in W$, construct v with $Tv = w$, OR show $\dim(\text{range } T) = \dim W$.
 - **Shortcut on finite-dim, equal-dimension spaces:** injective $\Leftrightarrow$ surjective $\Leftrightarrow$ invertible. Only need to prove **one**.
 - **"Maps a basis to a basis" shortcut:** if T sends a basis of V to a linearly independent list in W (with $\dim V = \dim W$), T is automatically invertible — check independence via a determinant or direct linear-combination argument.
 - **Existence/uniqueness of T given values on a basis:** cite the theorem directly — a linear map is completely and uniquely determined by its values on a basis. To justify "these conditions define a unique map," you usually just need to confirm the given list actually *is* a basis (check independence/length = dimension).
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4. Proving invariance (U is T -invariant)

Trigger: "show U is invariant under T ", "show $G(\lambda, T)$ is S -invariant"

1. Take arbitrary $u \in U$.
2. Show $Tu \in U$ using the definition of U (e.g., if $U = \text{null}(p(T))$, use that $p(T)$ and T commute).
3. Standard commuting trick: if $ST = TS$ and $U = \text{null}(q(T))$ or $\text{range}(q(T))$ for any polynomial q , then U is automatically S -invariant, because S and $q(T)$ commute too.

Always-invariant subspaces to remember (no proof needed, just cite): $\{0\}$, V , $\text{null}(T)$, $\text{range}(T)$, $\text{null}(p(T))$, $\text{range}(p(T))$ for any polynomial p .

5. Proving an "iff" / equivalence statement

Trigger: anything phrased "prove X if and only if Y" (e.g. Q4a on the practice final: inner product $\Leftrightarrow$ symmetric + PD)

1. Split into two directions explicitly: " $\Rightarrow$ " and " $\Leftarrow$ ". Don't try to do both at once.
 2. Forward direction: **assume the abstract property holds** (e.g. "assume $\langle \cdot, \cdot \rangle_A$ is an inner product"), then **extract concrete consequences** from each axiom one at a time (symmetry axiom $\Rightarrow$ matrix symmetric; positive-definiteness axiom $\Rightarrow$ matrix PD).
 3. Backward direction: **assume the concrete condition** (e.g. " A symmetric PD"), then **verify each axiom of the target definition** in turn, explicitly.
 4. Don't skip "trivial-looking" axioms like bilinearity — state them, even briefly.
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6. Finding an adjoint formula T^*

Trigger: "find the matrix/formula of T^* ", "compute $T^*(x^k)$ "

1. Write down what you're trying to prove is equal: guess the candidate S (via the standard-basis formula $T^* = \bar{A}^T$, or by symmetry of the problem), then confirm.
2. Compute $\langle Tv, w \rangle$ using the **given** inner product, expressed in terms of coordinates/coefficients.
3. Compute $\langle v, Sw \rangle$ the same way.
4. Match the two expressions termwise — this pins down S uniquely (adjoints are unique).
5. State clearly: "since this holds for all v, w , $T^* = S$."

Weighted/nonstandard inner product shortcut: if $\langle x, y \rangle_A = x^T A y$ (or $y^* A x$), then for $T_B(x) = Bx$: T_B^* has matrix $A^{-1} B^* A$. Derive it by matching $x^T B^T A y$ with $x^T A (C y)$, don't just quote it — the exam wants the derivation.

7. Proving self-adjoint / normal / PSD

Trigger: "show T is self-adjoint", "show ST is self-adjoint iff...", "show T is PSD"

- **Self-adjoint:** show $T^* = T$ directly (compute both sides in a basis), OR use $(ST)^* = T^* S^* = TS$ (if S, T self-adjoint) — then ST self-adjoint $\iff (ST)^* = ST \iff TS = ST$. This is the standard trick for "self-adjoint operators commute iff their product is self-adjoint" style questions.
 - **PSD:** pick the *easiest* equivalent characterization for the situation:
 - Structural (e.g. $T = A^* A$): use $\langle A^* A v, v \rangle = \langle A v, A v \rangle = \|A v\|^2 \geq 0$, plus $A^* A$ is self-adjoint since $(A^* A)^* = A^* A^{**} = A^* A$.
 - Eigenvalue-based: show self-adjoint (via spectral theorem) + all eigenvalues ≥ 0 .
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8. Diagonalizability arguments

Trigger: "is T diagonalizable", "determine the minimal polynomial and deduce diagonalizability"

Pick whichever is fastest given what you know:

1. **Eigenspace dimension count:** show $\dim E(\lambda_1) + \dots + \dim E(\lambda_m) = \dim V$.
2. **Minimal polynomial has no repeated roots** — often the fastest route once you've computed/guessed the min poly (e.g. via a polynomial identity like $(T - \lambda I)^2 = 0$).
3. **$\dim V$ distinct eigenvalues** — free diagonalizability, no further work.
4. **Self-adjoint/normal $\Rightarrow$ automatically diagonalizable** (spectral theorem) — don't do eigenspace work if this applies.

To find the minimal polynomial when given a matrix identity like $(C - 2I)^2 = 0$: translate the matrix identity into an operator identity ($(TC - 2I)^2 = 0$ acting on the whole space, not just one vector), confirm no smaller polynomial works by testing degree-1 candidates directly on a specific vector/matrix, then conclude by minimality.

9. Minimal polynomial *relative to a vector* / Krylov arguments

Trigger: "find p_v ", "prove $p_v \mid p$ "

1. $p_v \mid p$: since $p(T) = 0$ on all of V , in particular $p(T)v = 0$. By definition of p_v (smallest degree monic killing v), do polynomial long division $p = q p_v + r$ with $\deg r < \deg p_v$, plug in: $0 = p(T)v = q(T)p_v(T)v + r(T)v = r(T)v$. If $r \neq 0$, this contradicts minimality of p_v . So $r = 0$, i.e. $p_v \mid p$.
2. **Computing p_v directly for a diagonal-ish T** : find the eigenvalue components actually present in v 's expansion — p_v only needs to kill the eigenvalues that v actually has a nonzero component along, even if T has other eigenvalues elsewhere in V .

10. Generalized eigenspace / commuting-operator structure arguments

Trigger: " B commutes with A , determine block structure", " $G(\lambda, T)$ is S -invariant"

1. Cite: if $ST = TS$, $G(\lambda, T)$ is S -invariant (special case of the invariance template above, using $G(\lambda, T) = \text{null}((T - \lambda I)^{\dim V})$).
2. Decompose $V = G(\lambda_1, T) \oplus \dots \oplus G(\lambda_m, T)$.
3. Since S preserves **each** generalized eigenspace separately, the matrix of S (in a basis adapted to this decomposition) is **block diagonal**, with block sizes matching $\dim G(\lambda_i, T)$.
4. Within a block, if the corresponding block of T is a single Jordan block, use direct computation (solve $AB = BA$ entrywise for a general matrix B of that block size) to pin down exactly which matrices commute with it — typically upper-triangular Toeplitz-type matrices.

11. General proof-writing habits for this course

- **State which theorem/definition you're invoking before using it** — half credit is often for correctly identifying the right tool.
- **Show inclusion both ways** whenever proving set equality ($A = B$): prove $A \subseteq B$ and $B \subseteq A$ separately.
- **When stuck, count dimensions.** Rank-nullity and "sum of eigenspace dimensions $\leq \dim V$ " solve a surprising fraction of stuck proofs.
- **When a hint says "compute $(\cdot)^2(A)$ for arbitrary A "** — this is a "verify a polynomial identity holds on operators, not just on the original matrix" trick; expand directly and simplify, don't try to be clever.
- **Uniqueness proofs:** suppose two objects both satisfy the property, subtract/compare, show they must be equal.
- Keep an eye out for whether the problem is really over $\mathbb{R}$ or $\mathbb{C}$ — many facts (existence of eigenvalues, triangularizability) are drastically different between the two, and this course likes to test that distinction.

PART II — Additional Proof Templates (Broader Scope)

These extend beyond the patterns visible in past quizzes/exams, covering the rest of what's fair game given the syllabus. Templates 1-11 above stay as they are.

12. Proving eigenvalues exist (or don't)

Trigger: "show T has an eigenvalue", "show T has no real eigenvalue", "show T has a real eigenvalue"

- **Complex, finite-dim, $V \neq \{0\}$:** eigenvalues always exist — cite the theorem directly (proof idea if asked: take any nonzero v , the list $v, Tv, \dots, T^{\dim V} v$ has $\dim V + 1$ vectors so is dependent; the resulting polynomial relation factors over $\mathbb{C}$, and the first factor $(T - \lambda I)$ that fails to be injective on the relevant vector gives an eigenvalue).
- **Real, odd dimension:** always has a real eigenvalue — cite directly; don't re-derive unless asked.
- **Real, showing NO real eigenvalue exists:** exhibit an explicit 2-dimensional invariant subspace on which T acts as a rotation (no real eigenvector), or directly show $Tv = \lambda v \Rightarrow$ contradiction for every real λ by plugging into the specific formula for T .
- **Showing a specific number λ is/isn't an eigenvalue:** check whether $T - \lambda I$ is injective — either exhibit $v \neq 0$ with $(T - \lambda I)v = 0$ (is), or show $(T - \lambda I)v = 0 \Rightarrow v = 0$ (isn't).

13. Constructing an upper-triangularizing basis (induction)

Trigger: "show T can be put in upper-triangular form", "construct a basis making T upper triangular"

1. Base case: $\dim V = 1$ — any nonzero vector works trivially.
2. Inductive step: assume result for dimension $< n$. Since V is complex (or min poly splits), T has an eigenvalue λ ; let $U = \text{range}(T - \lambda I)$, which has $\dim U < \dim V$ (since $T - \lambda I$ isn't injective) and is T -invariant.
3. Apply the inductive hypothesis to $T|_U$ to get a basis $u_1, \dots, u_m$ of U making $T|_U$ upper triangular.
4. Extend to a basis of V by adding vectors $u_{m+1}, \dots, u_n$; this full list makes T upper triangular on V (any $v \in V$: $Tv = (T - \lambda I)v + \lambda v$, and $(T - \lambda I)v \in U = \text{span}(u_1, \dots, u_m)$).

14. Isometry / unitary equivalence-chain proofs

Trigger: "show S is an isometry iff...", "show these conditions on S are equivalent"

- These are proved as a **cycle** of implications ($1 \Rightarrow 2 \Rightarrow 3 \Rightarrow \dots \Rightarrow 1$), not all pairs separately — pick the easiest cycle order.
- Core algebraic move connecting norm-based and inner-product-based conditions: $\|Sv\|^2 = \langle Sv, Sv \rangle = \langle v, S^*Sv \rangle$, so $\|Sv\| = \|v\|$ for all $v \iff \langle v, (S^*S - I)v \rangle = 0$ for all $v \iff S^*S = I$ (uses that a self-adjoint operator equal to zero on all $\langle v, \cdot \rangle$ must be the zero operator).
- To go from " $S^*S = I$ " to "columns of $M(S)$ orthonormal": read off $\langle Se_i, Se_j \rangle =$ the (i, j) entry of $M(S)^*M(S)$, which is I exactly when columns are orthonormal.

15. SVD-existence / T^*T equivalence proofs

Trigger: "show these conditions (for existence of SVD) are equivalent"

- Central fact to lean on: $\text{null}(T^*T) = \text{null}(T)$ always — proof: $T^*Tv = 0 \Rightarrow \langle T^*Tv, v \rangle = 0 \Rightarrow \|Tv\|^2 = 0 \Rightarrow Tv = 0$; converse is immediate.
- Combine with rank-nullity applied to both T and T^*T (same domain) to get dimension-count equivalences (e.g. $\dim \text{range}(T) = \dim \text{range}(T^*T)$ follows immediately once null spaces match).
- $\text{range}(T^*T) = \text{range}(T^*)$: show $\subseteq$ directly (clear), then use a dimension count (via the null-space fact above + rank-nullity) to get equality.

16. Quotient spaces & induced maps (lighter, Week 1-2)

Trigger: "show $V / \text{null}(T)$ is isomorphic to $\text{range}(T)$ ", "show the induced map is injective"

1. Define $\tilde{T}(v + \text{null } T) = Tv$ — first confirm **well-defined**: if $v + \text{null } T = v' + \text{null } T$ then $v - v' \in \text{null}(T)$, so $T(v) = T(v')$, so the definition doesn't depend on the representative.
2. Injectivity of $\tilde{T}$: if $\tilde{T}(v + \text{null } T) = 0$ then $Tv = 0$ so $v \in \text{null}(T)$, so $v + \text{null } T = 0 + \text{null } T$, the zero coset.
3. Surjectivity onto $\text{range}(T)$: immediate from the definition.
4. Conclude isomorphism via dimension count if needed: $\dim(V / \text{null } T) = \dim V - \dim(\text{null } T) = \dim(\text{range } T)$ by rank-nullity.

17. Dual spaces, dual basis, annihilators (lighter, Week 1-2)

Trigger: "find the dual basis", "compute the annihilator", "show $\text{null}(T') = (\text{range } T)^0$ "

- **Dual basis coefficient extraction:** $\Phi_i(v) =$ the i -th coordinate of v in the basis $(v_1, \dots, v_n)$. For polynomial spaces with basis $1, x, \dots, x^n$: $\Phi_k(p) = p^{(k)}(0)/k!$ (derivative trick) — shift the evaluation point if the basis is centered elsewhere.
- **Annihilator proofs** $(U^0 = \{\varphi \in V' : \varphi(u) = 0 \ \forall u \in U\})$: to show $\varphi \in U^0$, just check φ vanishes on a **basis** of U (suffices by linearity).
- $\text{null}(T') = (\text{range } T)^0$: $\varphi \in \text{null}(T') \iff T'\varphi = 0 \iff \varphi(Tv) = 0 \ \forall v \iff \varphi$ vanishes on all of $\text{range}(T) \iff \varphi \in (\text{range } T)^0$. Same style for the other identity.

18. Polynomial-identity / Cayley-Hamilton arguments

Trigger: "show $q(T) = 0$ ", "use the factor theorem", "show T satisfies its characteristic polynomial"

- **Cayley-Hamilton shortcut:** if asked to show some specific polynomial relation $q(T) = 0$, first check whether q is (or is a multiple of) the characteristic polynomial — if so, cite Cayley-Hamilton directly instead of grinding through a computation.
- **Factor theorem chaining:** to show a polynomial with known roots $\lambda_1, \dots, \lambda_k$ factors completely, repeatedly pull out $(x - \lambda_i)$ factors one at a time via the factor theorem, reducing the degree by 1 each time.
- **Division algorithm arguments** (as in minimal-polynomial-divides-any-annihilating-polynomial proofs): write $q = s \cdot p + r$ with $\deg r < \deg p$, plug the relevant operator/vector in, use minimality of p to force $r = 0$.

19. Nilpotent operator proofs

Trigger: "show T is nilpotent", "show T restricted to $G(\lambda, T)$ is nilpotent"

- To show T nilpotent: show 0 is the **only** eigenvalue (sufficient over $\mathbb{C}$ — cite the theorem), or directly exhibit k with $T^k = 0$.
- To show $(T - \lambda I)|_{G(\lambda, T)}$ is nilpotent: every $v \in G(\lambda, T)$ satisfies $(T - \lambda I)^j v = 0$ for some $j \leq \dim V$ by definition of generalized eigenspace — so $(T - \lambda I)^{\dim V}$ kills all of $G(\lambda, T)$, i.e. is the zero operator restricted there.

20. Structured/explicit operator proofs (rank-one, block, weighted-inner-product)

Trigger: operator given by an explicit formula like $T(v) = av + b\langle v, u \rangle u$, or a specific small matrix

- Don't search for a slick general theorem — **just compute directly** on the pieces the formula naturally splits into (e.g. $\text{span}(u)$ and $u^\perp$ for rank-one perturbations; each Jordan block separately for block matrices).
- For rank-one type operators $T(v) = av + b \cdot \text{proj}_u(v)$: eigenvalue $a + b$ on $\text{span}(u)$, eigenvalue a (multiplicity $n - 1$) on $u^\perp$ — verify by direct substitution rather than re-deriving from scratch.
- When given a concrete matrix and asked to "determine all matrices commuting with it": set up B as a general matrix of the right size, expand $AB = BA$ entrywise, and solve the resulting linear system for the entries of B — this always works, even when no slicker argument is visible.

21. Commuting operators — deeper structural proofs

Trigger: "S and T commute, show they share a common eigenvector", "show they're simultaneously triangularizable"

- **Common eigenvector (complex, finite-dim, both operators nonzero space):** take any eigenvalue λ of T ; $E(\lambda, T)$ is S -invariant (commuting trick, template 4); restrict S to $E(\lambda, T)$ — since we're over $\mathbb{C}$, $S|_{E(\lambda, T)}$ has an eigenvalue, giving a vector that is simultaneously an eigenvector of S and T .
- **Simultaneous triangularization:** same idea as template 13, but at each inductive step choose the shared invariant subspace using the commuting-operator + eigenvalue trick above, rather than just one operator's eigenvalue.

22. Invertibility-without-eigenvalues arguments

Trigger: "show T is invertible" for a matrix where exact eigenvalues are hard to pin down

- **Gershgorin disks:** if every Gershgorin disk (centered at diagonal entry A_{ii} , radius = sum of absolute values of the rest of row i) excludes 0, then 0 is not an eigenvalue, so T is invertible. Fast when the matrix is diagonally dominant.
- **Direct injectivity check:** often faster than eigenvalues — assume $Tv = 0$, grind through the resulting linear system / functional equations, show $v = 0$ directly.

PART III — Computational Algorithms & Key Formulas

Pure procedure, no worked numbers — fill in your own numbers on scratch paper when practicing.

Characteristic polynomial

1. Pick any basis, write the matrix $M(T)$.
2. Compute $\det(M(T) - xI)$ by cofactor expansion (expand along a row/column with the most zeros).
3. For 2×2 : $x^2 - \text{tr}(M)x + \det(M)$ — shortcut, skip the full expansion.
4. Roots of this polynomial = eigenvalues; multiplicity of each root = algebraic multiplicity.

Eigenvalues & eigenvectors

1. Find eigenvalues via characteristic polynomial (above) or by inspection (triangular matrix: diagonal entries).
2. For each eigenvalue λ : solve $(M(T) - \lambda I)v = 0$ by row reduction; a basis for the solution space is a basis for $E(\lambda, T)$.

Diagonalization ($T = PDP^{-1}$)

1. Find all eigenvalues and a basis for each eigenspace (above).
2. Check $\sum \dim E(\lambda_i) = \dim V$ — if not, **not diagonalizable**, stop.
3. Let columns of P = the eigenvectors (in any order); D = diagonal matrix of corresponding eigenvalues in the same order.

Minimal polynomial (by hand)

1. Compute the characteristic polynomial and factor it: $(x - \lambda_1)^{a_1} \cdots (x - \lambda_m)^{a_m}$.
2. The minimal polynomial has the **same distinct roots**, each with exponent between 1 and a_i . Start by testing $(x - \lambda_1) \cdots (x - \lambda_m)$ (exponents all 1); if that annihilates T , done.
3. If not, increase exponents one at a time (smallest degree first) and test $q(T) = 0$ (apply $q(T)$ to each basis vector, or to a matrix directly) until it works.

Gram-Schmidt

Given linearly independent $(v_1, \dots, v_n)$, produce orthonormal $(e_1, \dots, e_n)$ spanning the same subspaces at each stage:

1. $e_1 = v_1 / \|v_1\|$.
2. For $k = 2, \dots, n$: subtract off the projection onto everything so far, $u_k = v_k - \sum_{j < k} \langle v_k, e_j \rangle e_j$, then normalize $e_k = u_k / \|u_k\|$.

QR factorization

1. Apply Gram-Schmidt to the columns of A to get orthonormal $q_1, \dots, q_n$ — these are the columns of Q .
2. R is upper triangular with entries $R_{jk} = \langle v_k, q_j \rangle$ for $j \leq k$ (read directly off the Gram-Schmidt computation).

Riesz representation vector

Given $\varphi : V \rightarrow \mathbb{R}$ linear:

1. Build an orthonormal basis $e_1, \dots, e_n$ (Gram-Schmidt if not already orthonormal).
2. Compute $\varphi(e_i)$ for each i .
3. $q = \sum_i \varphi(e_i) e_i$ (just $\sum \varphi(e_i) e_i$ over $\mathbb{R}$).

Adjoint matrix

- **Standard inner product:** $T^* \leftrightarrow$ conjugate transpose of $M(T)$.
- **Weighted inner product** $\langle x, y \rangle_A = y^* A x$ (A Hermitian PD): $T^* \leftrightarrow A^{-1} M(T)^* A$.
- **By hand for a nonstandard/given inner product:** write $\langle T v, w \rangle$ and $\langle v, S w \rangle$ both out explicitly using the given formula, match coefficients, solve for S .

PSD square root

1. Confirm $T = T^*$ and all eigenvalues ≥ 0 .
2. Orthogonally diagonalize: $T = P D P^{-1}$ with P having orthonormal eigenvector columns ($P^{-1} = P^*$).
3. $\sqrt{T} = P \sqrt{D} P^{-1}$, where $\sqrt{D}$ replaces each diagonal entry with its nonnegative square root.

SVD construction

1. Compute $T^* T$, find its eigenvalues $\sigma_1^2 \geq \dots \geq \sigma_m^2 > 0$ (positive ones only) and an orthonormal eigenbasis $e_1, \dots, e_m$ (extend to all of V if needed, but only the positive-eigenvalue ones matter).
2. Singular values $\sigma_i = \sqrt{\text{eigenvalue}}$.
3. Set $f_i = T e_i / \sigma_i$ — these come out orthonormal automatically.
4. $T v = \sum_i \sigma_i \langle v, e_i \rangle f_i$; matrix form $A = U \Sigma V^T$ with U 's columns = f_i , V 's columns = e_i .

Krylov subspace / minimal polynomial relative to a vector p_v

1. Compute $v, T v, T^2 v, \dots$ until the list becomes linearly dependent (i.e. $T^k v \in \text{span}(v, \dots, T^{k-1} v)$) — say this happens at $k = r$.
2. Write $T^r v = -(c_0 v + c_1 T v + \dots + c_{r-1} T^{r-1} v)$ from that dependence relation.
3. Then $p_v(x) = x^r + c_{r-1} x^{r-1} + \dots + c_0$.

Generalized eigenvector chains / Jordan basis (for a single Jordan block of size k at eigenvalue λ)

1. Find v with $(T - \lambda I)^{k-1} v \neq 0$ but $(T - \lambda I)^k v = 0$ (a generalized eigenvector of maximal order in that block).
2. The chain $v, (T - \lambda I)v, (T - \lambda I)^2 v, \dots, (T - \lambda I)^{k-1} v$ forms a basis for that Jordan block, ordered so that T sends each vector to $\lambda \cdot (\text{itself}) + (\text{previous vector in the chain})$.

Change of basis / matrix of a linear map

1. To find $M(T)$ w.r.t. basis $(v_1, \dots, v_n)$: compute $T v_i$ for each i , express the result in terms of $(v_1, \dots, v_n)$ — those coordinates form column i .
2. Change-of-basis matrix S (old $\rightarrow$ new): columns are the old basis vectors expressed in new-basis coordinates (or vice versa depending on convention) — new matrix = $S^{-1} M(T) S$ (similarity).